Predicting Customer Adoption Propensity Using Spending Behaviour

Final Year Project User Manual

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Table of Contents

1. Introduction

Welcome to the User Manual for the Customer Adoption Propensity Dashboard. This dashboard is a machine learning prototype developed for academic research purposes. Its primary goal is to predict the likelihood of a bank customer accepting a credit card offer based on their customer profile, financial behaviour, and campaign-related information. The dashboard is intended primarily for bank marketing officers and campaign decision-makers to support targeted credit card offer strategies.

By leveraging historical data, this tool assists marketing officers in identifying customers with higher adoption propensity, thereby supporting more targeted and informed marketing decisions.

⚠️ Important Scope Clarification

Please note that this system is developed strictly as a prototype for targeted marketing decision support. It is NOT:

- A credit approval system
- A credit scoring tool
- A loan eligibility predictor
- A fraud detection system
- A real banking deployment

2. Getting Started

2.1 Accessing the Dashboard

The dashboard is deployed on Streamlit Community Cloud, making it accessible from any modern web browser without the need for local installation.

Dashboard URL: https://customer-adoption-propensity.streamlit.app/

2.2 System Requirements

- A modern web browser (e.g., Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge).
- An active internet connection.
- No additional software or plugins are required.

3. Dashboard Navigation

The dashboard is structured into five main tabs, accessible via the top navigation bar. Each tab serves a distinct purpose in the analytical workflow.

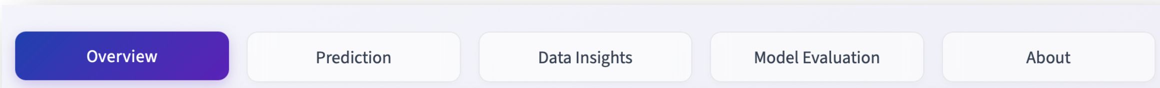

Figure 1: Dashboard Navigation Bar

3.1 Overview Tab

The landing page provides a high-level summary of the project. It includes key dataset metrics, simulated live analytics, and a brief description of the system's purpose. It acts as the command centre before diving into deeper analysis.

3.2 Prediction Tab

This is the core interactive module. Users can input specific customer characteristics (such as income level, bank accounts open, and quarterly balances) alongside campaign details (such as reward type). The system processes these inputs and outputs the predicted likelihood of the customer accepting the offer.

3.3 Data Insights Tab

This tab provides descriptive historical insights. Through interactive visualisations (such as pie charts and bar charts), users can explore the characteristics of customers who historically accepted or rejected offers. This helps in understanding the underlying data patterns.

3.4 Model Evaluation Tab

Designed for technical transparency, this tab presents the evaluation metrics of the machine learning models tested during development. It details why the final model was selected and displays its performance across various criteria.

3.5 About Tab

This section provides comprehensive documentation regarding the project's scope, intended users, system limitations, and an overview of the dataset used for training.

4. Performing a Prediction (Step-by-Step)

Follow these steps to generate a prediction using the dashboard.

Step 1: Navigate to the Prediction Tab

Click on the "Prediction" tab in the top navigation bar.

Step 2: Enter Customer Profile

Fill in the basic customer information:

- Income Level: Select from Low, Medium, or High.
- Credit Rating: Select from Low, Medium, or High.
- Household Size: Enter the number of people in the household.
- Own Your Home: Indicate Yes or No.
- Homes Owned: Enter the number of properties owned.

Step 3: Enter Banking Relationship & Campaign Details

- Bank Accounts Open: Number of current accounts held.
- Credit Cards Held: Number of existing credit cards.
- Overdraft Protection: Indicate Yes or No.
- Reward: Select the offer incentive (e.g., Air Miles, Cash Back, Points).
- Mailer Type: Select the marketing medium (e.g., Letter, Postcard).

Step 4: Enter Financial Behaviour

Enter the customer's quarterly balances for Q1, Q2, Q3, and Q4. The system will automatically calculate the Average Balance based on the non-zero inputs provided.

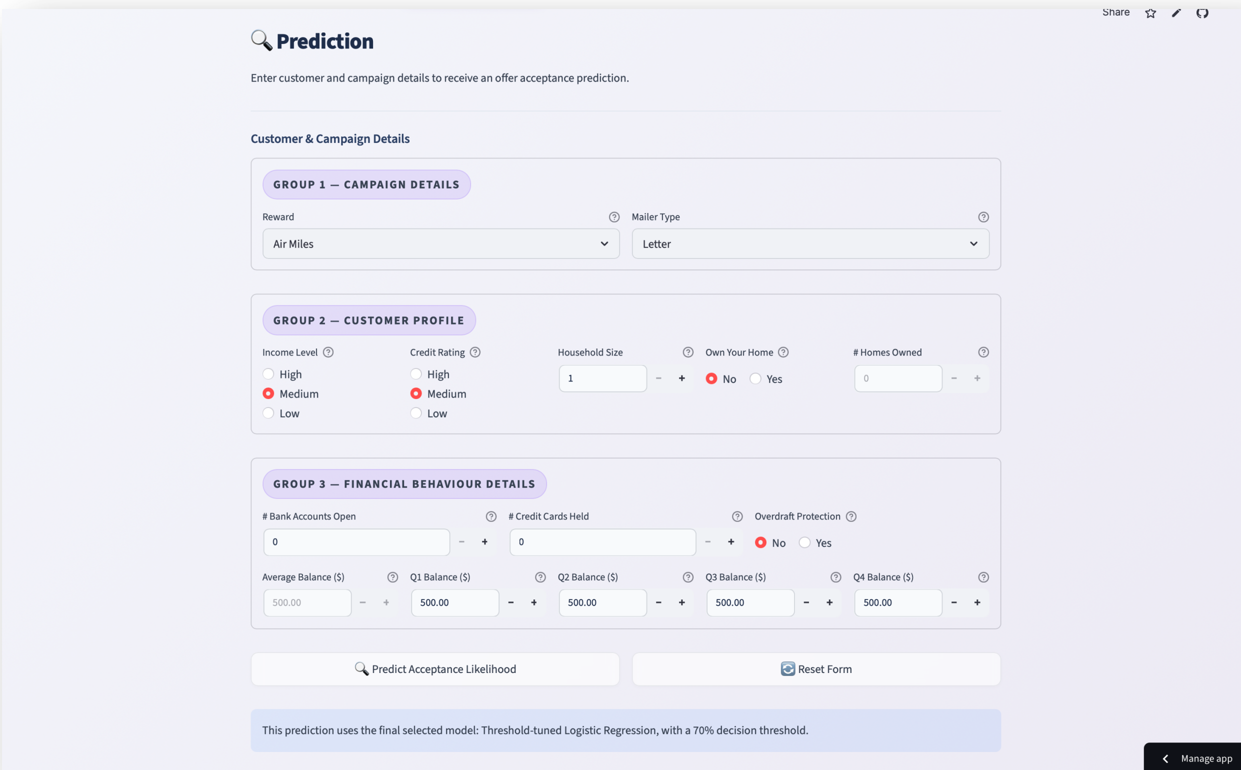

Figure 2: Data Entry in the Prediction Tab

Step 5: Generate Prediction

Click the "🔍 Predict Acceptance Likelihood" button at the bottom of the form.

Step 6: Interpret the Results

The system will display a results card showing:

1. Acceptance Probability: A percentage indicating the likelihood of the customer accepting the offer.
2. Marketing Recommendation: A priority tag based on the probability.
   - Recommended for Targeting (High probability)
   - Consider for Targeting (Medium probability)
   - Not Recommended for Targeting (Low probability)

💡 Resetting the Form

To clear your inputs and start over, click the "🔄 Reset Form" button. This will revert all fields to their default values.

5. Model Information

The dashboard is powered by a machine learning model trained on historical banking data. It does not learn in real-time; instead, it uses a static, highly optimised model to ensure consistent and reliable inferences.

5.1 Final Selected Model

The chosen model is a Threshold-tuned Logistic Regression. It was selected because it offered the best balance between detecting the minority class (actual adopters) and maintaining overall interpretability—a crucial factor in business environments.

5.2 Key Performance Metrics

Metric	Value	Interpretation
Decision Threshold	0.70	The probability cutoff used to classify a customer as a likely adopter. Tuned to reduce false positives.
ROC-AUC	0.764	Measures the model's overall ability to distinguish between adopters and non-adopters.
Accuracy	0.86	The overall proportion of correct predictions.
Recall (Minority Class)	0.42	The proportion of actual adopters that the model successfully identified.
F1-Score (Minority Class)	0.26	The harmonic mean of precision and recall for the minority class, indicating performance on the imbalanced dataset.

6. Limitations

Users must be aware of the following system limitations when interpreting results:

- Historical Bias: Predictions rely solely on historical patterns from the training dataset. Changes in economic conditions or customer behaviour over time may reduce accuracy.
- Imbalanced Data Constraint: Because offer acceptance is rare (~5.68% in the dataset), the model is optimised for recall. This means it may flag some customers who ultimately do not accept the offer (false positives) to ensure potential adopters are not missed.
- Static Nature: The deployed model does not continuously learn. It must be manually retrained offline with new data to remain relevant.
- Feature Limitation: The model only evaluates the features available in the input form. Unseen factors (e.g., a sudden life event for the customer) cannot be accounted for.

7. Troubleshooting

Dashboard is not loading or shows an error

- Cause: The Streamlit Cloud server might be asleep due to inactivity, or you have a poor internet connection.
- Solution: Refresh the web page. If the app has "gone to sleep," Streamlit will automatically wake it up, which may take up to a minute.

Average Balance is not calculating correctly

- Cause: The Average Balance is calculated strictly based on non-zero inputs in the Q1-Q4 fields.
- Solution: Ensure you have entered valid numerical values into the quarterly balance fields. The Average Balance field is disabled for manual entry and will update automatically as you type.

Prediction Result is missing

- Cause: The prediction button has not been clicked, or the form was reset.
- Solution: Ensure all fields are filled accurately and click the "Predict Acceptance Likelihood" button. If you switch tabs, you may need to re-click the predict button upon returning.

End of User Manual